

Total Numbers of Neurons and Glial Cells in Cortex and Basal Ganglia of Aged Brains with Down Syndrome—A Stereological Study

Anna Schou Karlsen and Bente Pakkenberg

Research Laboratory for Stereology and Neuroscience, Copenhagen University Hospital, Bispebjerg, DK-2400 Copenhagen NV, Denmark

Address correspondence to Anna Schou Karlsen, Research Laboratory for Stereology and Neuroscience, Copenhagen University Hospital, Bispebjerg, Bispebjerg Bakke 23, 11B2., DK-2400 Copenhagen NV, Denmark. Email: annakarsen@gmail.com.

The total numbers of neurons and glial cells in the neocortex and basal ganglia in adults with Down syndrome (DS) were estimated with design-based stereological methods, providing quantitative data on brains affected by delayed development and accelerated aging. Cell numbers, volume of regions, and densities of neurons and glial cell subtypes were estimated in brains from 4 female DS subjects (mean age 66 years) and 6 female controls (mean age 70 years). The DS subjects were estimated to have about 40% fewer neocortical neurons in total (11.1×10^9 vs. 17.8×10^9 , $2p \leq 0.001$) and almost 30% fewer neocortical glial cells with no overlap to controls (12.8×10^9 vs. 18.2×10^9 , $2p = 0.004$). In contrast, the total number of neurons in the basal ganglia was the same in the 2 groups, whereas the number of oligodendrocytes in the basal ganglia was reduced by almost 50% in DS (405×10^6 vs. 816×10^6 , $2p = 0.01$). We conclude that trisomy 21 affects cortical structures more than central gray matter emphasizing the differential impairment of brain development. Despite concomitant Alzheimer-like pathology, the neurodegenerative outcome in a DS brain deviates from common Alzheimer disease.

Keywords: aging, cell number, mental impairment, stereology, trisomy 21

Introduction

Down syndrome (DS) or trisomy 21 has a prevalence of approximately 1 in 700 live births and is associated with a number of deleterious phenotypes of which the most prominent are mental retardation, neonatal hypotonia, delayed development, and premature aging (see Contestabile et al. 2010). The life expectancy has improved considerably from 9 years of age in 1929 to 58 years of age in 2008 (Yang et al. 2002; Coppus et al. 2008). In adulthood, most DS subjects will have neurohistopathology resembling Alzheimer disease (AD) including amyloid plaques and neurofibrillary tangles, but the pathological processes for dementia in DS are characterized by a long prodromal period compared with AD (Wisniewski et al. 1985; Hof et al. 1995; Bondi et al. 2008). Two courses characterize the neuroanatomy related to DS: developmental abnormalities and progressive age-related changes. In general, growth retardation results in a small brain with a premature appearance including a rounded cerebrum and simple convolutional pattern, Figure 1. The sulci are superficial and reduced in number and the gyri are wide (except for the superior temporal gyrus, i.e., markedly narrow) (Wisniewski 1990; Delabar et al. 2006). The sizes of the basal ganglia and thalamus in young individuals with DS are comparable to those of the general population (Aylward et al. 1997; Pinter et al. 2001).

DS can be used as a model for investigating the structural brain changes resulting from progressive aging in human brains as well as the consequences of long-term neurodegenerative processes. A consistent finding in normal aging brains is the accumulation of iron in the form of nonheme iron as ferritin and hemosiderin, substances that induce oxidative stress by reaction with hydrogen peroxide, thereby disrupting lipid membranes, proteins, and deoxynucleic acids (see Stankiewicz and Brass 2009). Particularly structures that function in motor control such as the basal ganglia are affected (Bartzokis et al. 1994, 1997; Stankiewicz and Brass 2009). Magnetic resonance imaging of DS brains suggests a higher prevalence of iron deposition compared with the general population, and these deposits increase with age (Roth et al. 1996).

A previous study by Larsen et al. 2008, employed design-based stereology on brains from 3 males and 1 female DS fetuses and showed that by the 19th week of gestation, a reduction by 34% in the number of neocortical cells was obtained despite no gross morphological or density differences compared with normal brains. This is in agreement with another recent report that described decreased cell proliferation in the germinal matrix of fetuses with DS (17th to 21st gestational week) (Contestabile et al. 2007).

We estimated the total number of neocortical neurons and glial cells in 4 aged female subjects with DS to see if the smaller cell numbers in the fetal brain of DS subjects are also found later in life. In addition, the numbers of neurons and glial cells in the basal ganglia were estimated to compare cortical and subcortical cell populations in both groups.

Materials and Methods

Patient Characteristics and Tissue Processing

Hemispheres from 4 female DS subjects (mean age 66 years, range 61–70 years) were kindly provided by the Netherlands Brain Bank (NBB) and 6 female controls (mean age 70 years, range 60–80 years) were obtained postmortem according to Danish law regarding autopsied human tissue. For clinical information on cause of death, postmortem interval before fixation, and fixed hemisphere weight, see Table 1. The brains were fixed for at least 5 months in 0.1 M sodium phosphate-buffered 4% formaldehyde (pH 7.2). The cerebrum was detached from the brainstem at the level of the third cranial nerve, and the meninges were removed. The hemispheres were sliced coronally into 3 cm thick slabs starting from a random position within the first 3 cm. Each slab was embedded in paraffin using the Leica ASP300 S tissue processor followed by exhaustive coronal sectioning on a Leica SM2400 sliding microtome with a setting of 40 μ m. The sections were mounted on Superfrost Plus microscope slides (Thermo Scientific) and dried.

The neuropathology of the patients with DS was consistent with regard to the presence of numerous widespread senile plaques and neurofibrillary tangles in a disrupted interneuronal network in the

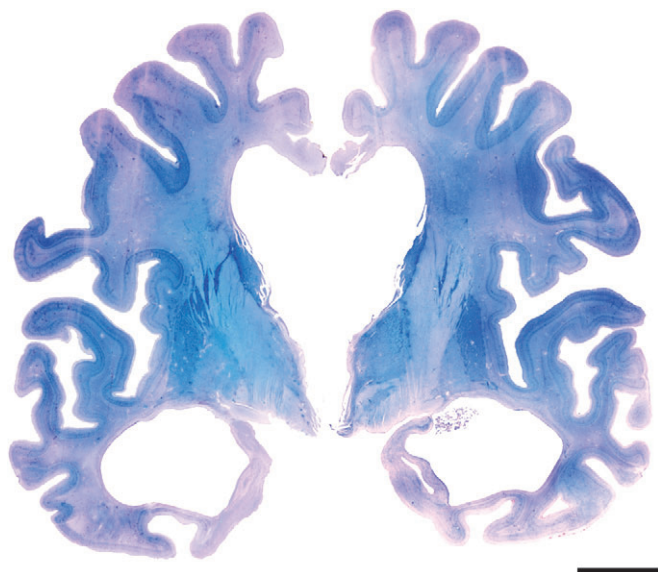


Figure 1. Nissl-stained coronal section from a 70-year-old DS subject (formalin fixed, paraffin embedded). The bar indicates 1 cm.

Table 1
Clinical details of the individuals

	Age	Height (cm)	Hemisphere (gram)	Country	Cause of death	PMD (h)
Controls	80	162	—	DK	Incomp. cord	12:00
	74	160	469	DK	AMI	48:00
	64	154	—	DK	AMI	24:00
	60	161	377	DK	AMI	48:00
	74	152	389	DK	Hemopericardium	10:30
	70	169	482	DK	AMI	04:00
Mean	70.3	159.7	429.3			
DS	61	—	244	NED	Pneumonia	05:30
	64	148	231	NED	Pneumonia	04:00
	70	162	245	NED	Ileus	09:00
	70	—	290	US	—	06:00
Mean	66.3	159.7	252.5			

Note: PMD, postmortem delay; Incomp. cord., incompensatio cordis; AMI, acute myocardial infarction; DK, Denmark; NED, The Netherlands; US, United States.

frontal and temporal cortices. None of the patients had any reports of hemochromatosis.

A fraction of the sections (see Stereological Design) was dewaxed and hydrated followed by Nissl-staining with Giemsa's azur eosin methylene blue solution (Merck) and KHPO_4 (67 mmol/L, pH 4.5) in a 1:4 ratio. Perls stain (Biostain Ready Reagents Ltd.) was used to detect ferric salts in the tissue sections. Following instructions from the manufacturer, the sections were dewaxed and stained with Perls combined with Neutral Red (0.5% Aq).

Stereological Design

The sections were sampled using systematic uniform random sampling (Gundersen et al. 1999). The sampling sequence for neocortical estimates aimed at sampling approximately 18–26 sections per specimen with a sampling fraction ranging from 1/125 to 1/200 (depending on the length of the specimen) and 8 to 10 sections for the basal ganglia volume (sampling fraction 1/100–1/150). The sampling of disectors for counting cells was performed using a predetermined fraction of the area of interest and in a known fraction of the section thickness. In essence, this design was thus just as well a fractionator method as a density/volume combination. The density/volume combination was preferred to also obtain numerical density estimates and volume estimates.

Estimation of Volume, Surface, and Thickness

Estimates of regional and total brain volumes were obtained by employing Cavalieri's principle counting points applied randomly to cross-sectional areas. The 6-layered cortex defined neocortical regions. Included in the volume estimates for allocortex were the 3-layered archicortex and paleocortex as well as the transitional mesocortical areas, the cingulate gyrus, uncus, subiculum, and the entorhinal cortex (whole). The basal ganglia, amygdala, and thalamic nuclei represented central gray matter. Included in the basal ganglia estimations (volume and cell numbers) were the striatum, globus pallidum, nucleus accumbens, and subthalamic nuclei, which were all highly reactive for ferritin.

The brain volumes were found by multiplying the total number of counted points in the regions of interest by the area per test point and by the average distance between the sections (Gundersen and Jensen 1987; Gundersen et al. 1999). Cortical surface area was estimated in order to derive the cortical thickness. When estimating surface areas, test line probes (uniform and random) were applied to parallel, frontal sections (Gundersen 1985), and the estimates were thus not unbiased. However, the actual bias was previously measured in human neocortex and is so small ($\approx 3\%$) that it can be ignored (Oster et al. 1993).

Due to substantial tissue shrinkage that is inevitable when embedding in paraffin, the volume and surface estimates are multiplied by shrinkage correction factors assuming that the shrinkage is uniform and concentric (see Uylings et al. 1986).

The volumetric shrinkage correction factor is calculated as the inverse of the residual volume after shrinkage as $V_{\text{corr}} = \frac{1}{1 - \frac{V_{\text{paraffin}}}{V_{\text{formalin}}}}$.

To calculate the surface shrinkage factor, the linear shrinkage is found as $L_{\text{shrinkage}} = 1 - \sqrt[3]{V_{\text{paraffin}}/V_{\text{formalin}}}$ and the surface shrinkage correction factor as $S_{\text{corr}} = 1 / L_{\text{shrinkage}}^2$.

The control brains had a mean shrinkage of 45.8% (0.071) and the DS brains 48.0% (0.066). This difference was not statistically significant ($2p = 0.33$), so we used the mean shrinkage of all brains of 46.7% (0.068) for calculating the volume correction factor.

It should be noted that differential shrinkage of brain tissue can never be excluded; however, estimates of the total numbers of cells are unaffected by shrinkage artifacts.

The mean cortical thickness was estimated as the volume of the cortex divided by the surface area, both corrected for shrinkage.

Cell Density and Number Estimation

The numerical cell densities of neurons and glial cells were estimated with an optical disector (Gundersen 1986). The sections were analyzed on a BX-50 Olympus microscope equipped with a Heidenhain electronic microcator with a digital readout (for measurements in the z -direction) with a precision of 0.5 μm and an X - Y - Z motorized stage (PRIOR Scientific). With a high-resolution color camera (JAI CV-3200) mounted on the microscope and a 2.5 $\times$ photo eyepiece (Olympus), live images were forwarded to the computer running the CAST Stereology software package (v. 2.1.5.8, Visiopharm). Sampling in the z -axis was done with a disector height of 20 μm and upper and lower guard zones of 7 μm . Neurons and glial cells were counted in one defined region, ΣQ , and the numerical density, N_V , estimated (Pelvig et al. 2008). The total number of particles was estimated as the product of the reference volume (uncorrected for shrinkage) and the numerical density in the samples. The bilateral total number was obtained by multiplying by 2.

Differentiation of the cell types was based on cell morphology according to the criteria listed in Pelvig et al. (2008). In general, neurons have larger nuclei with darkly stained nucleoli and visible cytoplasm. Oligodendrocytes are small round and contain dense heterochromatin and have no visible cytoplasm, whereas astrocytes are paler with heterochromatin situated in granules and a sharp profiled nuclear membrane. The microglia have elongated or comma-shaped nuclei with dense peripheral chromatin.

The precision of estimates was expressed as the coefficient of error (CE) (Gundersen et al. 1999). The observed interindividual coefficient of variation (CV = standard deviation/mean) is reported in parenthesis after the group means in the Results section. For the Cavalieri volume estimation, an average of 1271 (range 733–2081) points was counted

per brain corresponding to total neocortical volumes being estimated with CE of less than 1%.

For the density estimations, an average of 971 neurons (range 683–1453) was counted per brain and an average of 970 glial cells (range 716–1759) (i.e., averages of 552 oligodendrocytes, 292 astrocytes, and 126 microglia), which gives a CE of about 1–2%.

From a methodological point of view, these CE values were low compared with the magnitude of the biological variation (~20%). However, this was equal to a CE(Q) of 6–7% for cell counting per region (on average 243 neurons (range 148–418) and 242 glial cells (range 152–493)) and a CE for point-counting with a CE(P) of 4% per region. The cells were counted in an average 119 disectors (range 54–184) per region.

Differences between groups were tested with an unpaired Student's t -test with a significance level of 0.05 (2-tailed). If normality or equal variance tests failed, a nonparametric Mann-Whitney Rank Sum Test was employed. Due to the power-reducing effect of a Bonferroni correction in relation to the small sample size of this study, the correction was omitted.

Results

Estimated quantities of the major brain compartments are shown in Figure 2.

The brains from patients with DS had an average of 10.9×10^9 neocortical neurons (range 9.50 – 12.0×10^9 (0.10)), whereas the controls had a total of 17.9×10^9 neocortical neurons (range 15.2 – 21.1×10^9 (0.12)). The difference of 7.00×10^9 cells (95% confidence interval [CI] of $\pm 2.64 \times 10^9$) between the 2 groups was highly statistically significant ($t = 6.08$ with degrees of freedom [df] = 8 at $2p \leq 0.001$).

A difference of 5.68×10^9 (95% CI of $\pm 3.51 \times 10^9$) was found in the total number of glial cells. The DS brains on average had 12.5×10^9 cells (range 11.5 – 13.5×10^9 (0.08)) and the controls had 18.2×10^9 cells (range 15.0 – 21.8×10^9 (0.14)) ($t = 3.8$ with df = 8 at $2p = 0.005$). The glial cells were subdivided into oligodendrocytes, astrocytes, and microglia. From these data, the DS glial distribution was 53.5% oligodendrocytes, 34.8% astrocytes, and 11.3% microglia, whereas the controls had 57.1% oligodendrocytes, 29.1% astrocytes, and 13.6% microglia. There was no difference in the global glial/neuron ratio in the 2 groups with 1.15 (0.05) in DS and 1.03 (0.15) in controls ($t = -1.37$ at $2p = 0.21$).

The global neuronal density was not different between the 2 groups with $86.9 \times 10^6/\text{cm}^3$ (0.05) in DS (range 80.6 – $92.0 \times 10^6/\text{cm}^3$) and $82.2 \times 10^6/\text{cm}^3$ (0.09) in controls (range 70.9 – $92.4 \times 10^6/\text{cm}^3$) ($t = -0.96$ with df = 7 at $p = 0.37$). This was also the case for cortical subregions except the occipital cortex, in which the DS group had significantly lower neuronal density compared with controls: $108 \times 10^6/\text{cm}^3$ (0.05) versus $126 \times 10^6/\text{cm}^3$ (0.10), respectively ($t = 2.63$ with df = 7 at $2p = 0.034$). The global glial cell density did not differ significantly, with $105 \times 10^6/\text{cm}^3$ (0.04) in DS and $88.6 \times 10^6/\text{cm}^3$ (0.03) in controls (Mann-Whitney Rank Sum: $2p = 0.38$). However, in the temporal cortex, the glial cell density was significantly increased in DS brains compared with controls: $121 \times 10^6/\text{cm}^3$ (0.08) versus $88.6 \times 10^6/\text{cm}^3$ (0.07) ($t = -2.52$ with df = 7 at $2p = 0.04$), respectively. A significant increase in the density of astrocytes in the temporal cortex was found: $85.9 \times 10^6/\text{cm}^3$ (0.13) in DS versus $46.7 \times 10^6/\text{cm}^3$ (0.12) in controls (Mann-Whitney Rank Sum Test $p = 0.02$).

Volume estimates (shrinkage corrected) are listed in Table 2. Estimated global volumes of neocortex in the 2 groups differed significantly: 230 cm^3 (0.13) in DS and 403 cm^3 (0.17) in controls ($t = 4.10$ with df = 7 at $2p = 0.005$). This reduction was manifested locally with reductions of 44.4, 45.6, and 45.5% for frontal, temporal, and parietal cortices, respectively. The occipital cortex in the DS brains did not differ significantly from the control group ($t = 1.54$ with df = 7 at $2p = 0.17$).

The volume of the ventricles (lateral and third) was significantly larger (70.0 cm^3 (0.17)) than that of the controls (20.5 cm^3 (0.74)) (Mann-Whitney Rank Sum: $2p = 0.02$). Significant differences were also found in the archicortex (19.7 cm^3 (DS) vs. 33.2 cm^3 (control) (0.19)) ($t = 3.22$ with df = 7 at $2p = 0.02$) and in the central gray regions (23.0 cm^3 (DS) (0.05) vs. 37.9 cm^3 (control) (0.16)) ($t = 3.89$ with df = 7 at $2p = 0.006$). The significant difference in white matter volume of 50.2 (207 cm^3 (0.12)) in DS vs. 416 cm^3 (0.13) in controls ($t = 5.83$ with df = 7 at $2p \leq 0.001$) between DS and controls was more prominent than the total neocortical volume reduction of 43.0%. The volume estimations suggested lower variability among the DS group compared with the control group.

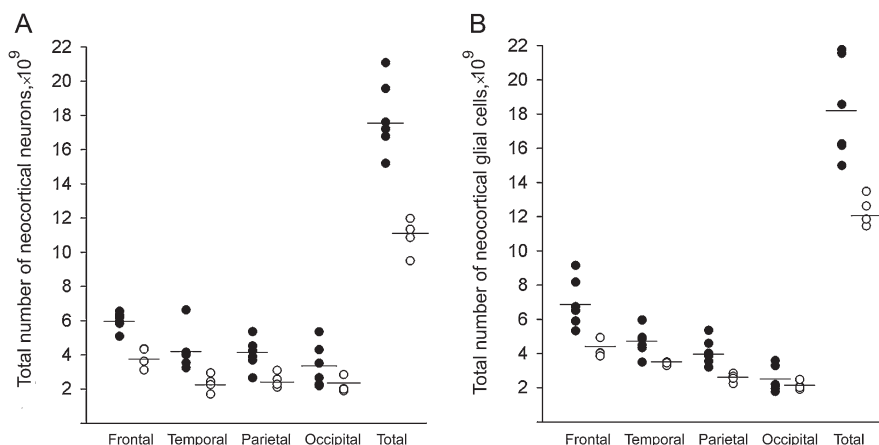


Figure 2. Total number of neurons (A) and glial cells (B) in the 4 major neocortical regions in DS (○) and controls (●). Each point represents one brain and is the product of the cellular density and volume of the region of interest. Horizontal bars indicate group mean. (A) Statistically significant differences in the number of neurons for the total neocortex and frontal cortex were $\alpha < 0.001$ and for the temporal and parietal cortices were $\alpha < 0.05$. The difference in the occipital cortex was not significant. (B) Statistically significant differences in the estimated number of glial cells in the total neocortex and frontal, temporal, and parietal cortices were $\alpha < 0.05$. The difference in the occipital cortex was not significant.

The entire neocortical surface area in the controls ranged from 1341 to 1963 cm² (mean 1572 cm² (0.14)), which was significantly different from the surface area estimated in DS, which ranged from 895 to 1050 cm² (mean 950 cm² (0.09)) ($t = 3.99$ with $df = 7$ at $2p = 0.005$) (Table 2—corrected for shrinkage).

The global neocortical thickness did not differ significantly between the 2 groups: 2.43 mm (0.05) in DS brains and 2.56 mm (0.06) in controls ($t = 1.03$ with $df = 7$ at $2p = 1.24$). This observation was also true for each cortical subregion (Table 2).

Basal Ganglia

Marked iron deposits were detected in the basal ganglia (striatum, globus pallidus, subthalamic nucleus, nucleus accumbens) of all 4 DS brains. The internal globus pallidus showed the most intense ferritin staining. The estimated numbers of neurons and glial cell subtypes in the basal ganglia are plotted in Figure 3. The DS brains were estimated to have an average of 373×10^6 (range $336\text{--}408 \times 10^6$ (0.09)) neurons and the controls 415×10^6 (range $289\text{--}504 \times 10^6$ (0.24)) neurons in the basal ganglia (Mann-Whitney Rank Sum: $2p = 0.69$). Likewise, there was no significant difference in the number of astrocytes: 231×10^6 (0.14) in DS basal ganglia and 275×10^6 (0.26) in controls ($t = 1.11$ with $df = 6$ at $2p = 0.31$). In contrast, the estimated number of oligodendrocytes differed by 50%: 405×10^6 (0.24) in DS and 816×10^6 (0.27) in controls ($t = 3.40$ with $df = 6$ at $2p = 0.01$). This was also evident from the estimated oligodendrocyte density, which was decreased by 33% with $54.0 \times 10^6/\text{cm}^3$ (0.34) in DS and $80.5 \times 10^6/\text{cm}^3$ (0.11) in controls ($t = 2.56$ with $df = 6$ at $2p = 0.04$). In addition, the number of microglia was also significantly different: 78.5×10^6 (0.20) in DS and 129×10^6 (0.40) in controls ($t = 6.27$ with $df = 6$ at $2p \leq 0.001$).

The estimated bilateral volume (uncorrected) of the basal ganglia was 6.98 cm^3 in DS (range $5.79\text{--}8.58 \text{ cm}^3$ (0.17)) and 10.1 cm^3 in the controls (range $8.22\text{--}12.0 \text{ cm}^3$ (0.20)), a difference of about 30% that was not statistically significant (Mann-Whitney Rank Sum: $2p = 0.11$). The neuronal density of $54.8 \times 10^6/\text{cm}^3$ (0.21) in the DS basal ganglia was not statistically different compared with the $41.0 \times 10^6/\text{cm}^3$ (0.10) in the controls ($t = -2.24$ with $df = 6$ at $2p = 0.07$).

Discussion

Aged women with DS were estimated to have almost 40% fewer neocortical neurons and almost 30% fewer glial cells than age-matched controls with no overlap between groups. This reduced number of neocortical cells correlates with a substantially smaller neocortical volume with no changes in the cell densities. This finding contrast what has previously been assumed for DS brains based on density estimates obtained by semiquantitative methods (Ross et al. 1984). The estimated number of neocortical neurons for the subjects with DS that ranges from 9.50 to 12×10^9 is, to our knowledge, the lowest estimated neocortical neuron number obtained so far in adult human brains. The limit for maintaining a normal intelligent life is not known but the range for normal people is $15\text{--}32 \times 10^9$ neurons (Pakkenberg and Gundersen 1997).

The reduced number of neocortical cells in brains of adults with DS may indicate that developmental deficits evident as early as the 19th gestational week constitutes a permanent

Table 2

Mean bilateral volumes of major structural components (cm³), surface areas (cm²), and cortical thicknesses (mm) in DS and control brains (corrected for shrinkage induced by tissue processing)

Region		Volume CV	2p df = 7	Surface CV	2p df = 7	Thickness CV	2p df = 7
Frontal	Control	157 (0.20)	0.01	596 (0.18)	0.007	2.62 (0.05)	NS
	DS	87.6 (0.17)		363 (0.14)		2.40 (0.09)	
Temporal	Control	105 (0.20)	0.01	433 (0.19)	0.02	2.44 (0.12)	NS
	DS	57.1 (0.22)		240 (0.17)		2.38 (0.07)	
Parietal	Control	91.8 (0.18)	0.005	336 (0.21)	0.02	2.76 (0.14)	NS
	DS	50.0 (0.11)		203 (0.06)		2.47 (0.17)	
Occipital	Control	48.5 (0.28)	NS	207 (0.26)	NS	2.35 (0.17)	NS
	DS	35.7 (0.09)		143 (0.25)		2.57 (0.18)	
Total neocortex	Control	403 (0.17)	0.005	1572 (0.17)	0.005	2.56 (0.06)	NS
	DS	230 (0.13)		950 (0.13)		2.43 (0.09)	
Allocortex	Control	33.2 (0.19)	0.02				
	DS	19.7 (0.06)					
Central gray nuclei	Control	37.9 (0.16)	0.006				
	DS	23.0 (0.05)					
White matter	Control	416 (0.13)	<0.001				
	DS	207 (0.12)					
Ventricles	Control	20.5 (0.27)	0.02				
	DS	70.0 (0.17)					

CV = standard deviation/mean, NS = not significant.

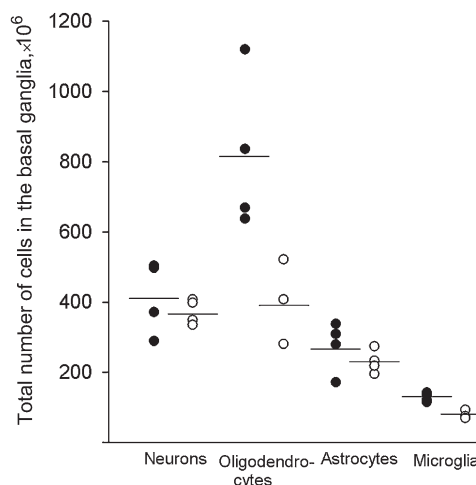


Figure 3. Total number of neurons and glial cell subtypes in the basal ganglia in DS ($n = 4$) (○) and controls ($n = 4$) (●). Each point represents one brain and is the product of the cellular density and volume of the region. Horizontal bars indicate group mean. A statistically significant difference was detected in the number of oligodendrocytes and microglia at $\alpha < 0.05$. The differences in the numbers of neurons and astrocytes were not significant.

condition throughout life. Evidently we cannot exclude a compensational increase in early adulthood followed by an age-related reduction later in life given that we only have access to 2 time frames. However, imaging studies have shown that adolescent and young adults with DS have smaller brain volumes compared with age-matched healthy subjects (Raz et al. 1995; Pinter et al. 2001), and we therefore find it more likely that the total number of neocortical cells never reach normal levels.

Another limitation to our study is the low sample size. However, the differences in our results are highly statistically significant; for example, a retrospective analysis of the power of the statistical test based on the 40% reduction in the number of neocortical neurons gave a power of 0.99. This ensured that the 4 subjects in the DS group and 6 controls were enough to detect the differences reported here.

If the lower number in the adult neocortex of DS is permanent, it is likely the consequence of uncompensated developmental deficits evident as early as the 19th gestational week at which time there are 34% fewer neocortical cells in the forebrain as compared with normal brains as reported by Larsen et al. (2008). Recent research suggests that the delayed cell cycle detected in both the hippocampus and the ventricular germinal matrix of DS fetuses is a main contributor to the reduced neurogenesis (Contestabile et al. 2007). The final number of neurons in the mature neocortex is partially determined by the number and proliferation of progenitor cells in the ventricular zone present at the beginning of neurogenesis (Kornack and Rakic 1998). Concerning the relatively unaffected number of cells in the basal ganglia, one explanation may be that the ganglionic eminence, which provides neuronal cells to deep brain nuclei, persists up to 1 year after birth and therefore has more time to compensate for the initial impairment reported by Sweeney et al. (1989). This is in contrast to the ventricular zone, which disappears shortly after completion of neuronal migration (see Encha-Razavi and Sonigo 2003).

Cortical thickness is one of the most stable parameters of the human brain, and it exhibits low biological variation in contrast to the cerebral volume and surface (Pakkenberg and Gundersen 1997). AD is the only condition reported so far that leads to significant thinning of the neocortex despite a normal neocortical neuron number (Regeur et al. 1994; Bundgaard et al. 2001; Pelvig et al. 2003). This is in contrast to DS, where the cortex thickness is normal and the relative interneuronal distances presumably intact despite the AD-related neuropathology. According to the radial-unit hypothesis suggested by Rakic, it is mainly the number of progenitor cells prior to neurogenesis that establishes the future number of radial columnar units forming the neocortex (Rakic 1995). The number of radial units determines the volume and surface of the neocortex, wherefore fewer progenitor cells as a consequence of and in combination with delayed proliferation will lead to smaller brains but not necessarily to thinner cortices.

Despite a significant reduction in the total central gray volume (including amygdala, thalamic nuclei, and basal ganglia) in the DS brains, only a small reduction was observed when estimating selectively the basal ganglia volume, where the number of neurons was almost the same in the 2 groups due to a slight increase in cell density. This is in line with in vivo imaging studies on young subjects with DS, where volume estimates of the basal ganglia and thalamus show no deviations from the general population (Aylward et al. 1997; Pinter et al. 2001). The glial cell counts in the basal ganglia showed that the number of astrocytes was similar in DS and controls, whereas the numbers of oligodendrocytes and microglia were significantly smaller in the DS brains. We hypothesize that the difference in the number of oligodendrocytes is a loss due to the coincident iron overload in this glial cell subtype, since in vitro studies have showed that oligodendrocytes are highly susceptible to iron-induced toxicity via oxidative stress (Kress et al. 2002; Zhang et al. 2005). Considering the importance of the basal ganglia for proper motor function, one could speculate that the loss of oligodendrocytes may be related to the deterioration of motor function that is ubiquitous in aged patients with DS.

In conclusion, all brains had severe neuropathological changes as seen in AD, but since it has been shown that AD

is not followed by reductions of the total number of neuron or glial cells in neocortex, it is not likely that the smaller nerve cell number in DS is caused by the concomitant Alzheimer pathology. In volume and surface area the brain of DS is simply a normal brain scaled down by 30–40%, while the height of the neocortex and the nerve and glial cell densities are within normal limits. This is in contrast to AD, where the cortical cell density is increased, while the thickness of the cortex is significantly reduced. Thus, although DS and AD share resemblance in neuropathology they differ in neocortical cell numbers, cell density, and cortical thickness. Finally, we have shown that the critical lowest number of neocortical neurons necessary to preserve and conduct normal intelligence is around 15 billion neurons and that a reduction below this limit of just a few billion neurons may make the difference in normal brain function and mental retardation.

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Notes

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